DSA & Advanced DSA Roadmap

Phase 1 — Arrays

Learning Objectives

By the end of this phase, learners will be able to:

- Understand how arrays are stored in memory.
- Explain static and dynamic arrays.
- Implement common array operations.
- Analyze time and space complexity of array operations.
- Identify and apply array-based problem-solving patterns.
- Solve Easy to Medium level array problems independently.

Implementation & Internals

Core Concepts

- What is an Array?
- Static vs Dynamic Arrays
- Memory Layout of Arrays
- Index-Based Access
- Array Traversal
- Insertion Operations
- Deletion Operations
- Resizing Strategy
- Dynamic Array Internals
- Amortized Analysis

Implementation Exercises

- Build a static array wrapper with bounds checking.
- Build a dynamic array (resizable array) from scratch, including a doubling resize strategy.
- Implement insert-at-index and delete-at-index, and measure their cost.
- Implement and time amortized append across repeated doublings.

Pattern Mastery

Pattern 1: Traversal

Concepts

- Linear Scan
- Forward Traversal

- Reverse Traversal
- Multiple Traversals

Practice Problems

- Largest Element in Array
- Second Largest Element
- Check Sorted Array
- Remove Duplicates from Sorted Array
- Leaders in Array

LeetCode

  - https://leetcode.com/problems/find-pivot-index/
  - https://leetcode.com/problems/running-sum-of-1d-array/

Pattern 2: Frequency Counting

Concepts

- Counting Occurrences
- Frequency Arrays
- Frequency Maps

Practice Problems

- Frequency of Elements
- Majority Element
- Missing Number
- Duplicate Number

LeetCode

  - https://leetcode.com/problems/contains-duplicate/
  - https://leetcode.com/problems/majority-element/

Pattern 3: Prefix Sum

Concepts

- Prefix Array Construction
- Range Sum Queries
- Prefix Optimization

Practice Problems

- Range Sum Query
- Equilibrium Index
- Subarray Sum

LeetCode

  - https://leetcode.com/problems/range-sum-query-immutable/
  - https://leetcode.com/problems/subarray-sum-equals-k/

Pattern 4: Difference Array

Concepts

- Range Updates
- Lazy Range Modification
- Query Optimization

Practice Problems

- Range Addition
- Corporate Flight Bookings
- Interval Increment Queries

LeetCode

- https://leetcode.com/problems/corporate-flight-bookings/

Pattern 5: Two Pointers

Concepts

- Opposite Direction Pointers
- Same Direction Pointers

Practice Problems

- Pair Sum
- Remove Duplicates
- Move Zeroes
- Container With Most Water

LeetCode

- https://leetcode.com/problems/two-sum-ii-input-array-is-sorted/
- https://leetcode.com/problems/move-zeroes/
- https://leetcode.com/problems/container-with-most-water/

Pattern 6: Sliding Window

Concepts

- Fixed Window
- Variable Window

Practice Problems

- Maximum Sum Subarray
- Longest Unique Subarray
- Maximum Consecutive Ones

LeetCode

- https://leetcode.com/problems/maximum-average-subarray-i/
- https://leetcode.com/problems/permutation-in-string/
- https://leetcode.com/problems/longest-repeating-character-replacement/

Pattern 7: Kadane's Algorithm

Concepts

- Maximum Subarray
- Running Sum Optimization

Practice Problems

- Maximum Subarray Sum
- Maximum Circular Subarray

LeetCode

- https://leetcode.com/problems/maximum-subarray/
- https://leetcode.com/problems/maximum-sum-circular-subarray/

Pattern 8: Merge Intervals

Concepts

- Interval Overlap
- Interval Consolidation

Practice Problems

- Merge Intervals
- Insert Interval
- Meeting Rooms

LeetCode

- https://leetcode.com/problems/merge-intervals/
- https://leetcode.com/problems/insert-interval/

Note: "Meeting Rooms" (LeetCode 252) requires LeetCode Premium. Free alternative: implement the can-attend-all-meetings check directly (sort by start time, verify no adjacent overlap) as a non-graded exercise.

Pattern 9: Cyclic Sort

Concepts

- Index Placement Strategy
- Missing Number Problems

Practice Problems

- Missing Number
- First Missing Positive
- Find Duplicate Number

LeetCode

- https://leetcode.com/problems/missing-number/
- https://leetcode.com/problems/first-missing-positive/
- https://leetcode.com/problems/find-the-duplicate-number/

Phase Assessment

Implementation Tasks

- Build Dynamic Array from Scratch
- Implement Insert/Delete Operations
- Analyze Complexity

Recommended LeetCode Target

Easy Problems

- https://leetcode.com/problems/contains-duplicate/
- https://leetcode.com/problems/move-zeroes/
- https://leetcode.com/problems/majority-element/
- https://leetcode.com/problems/missing-number/
- https://leetcode.com/problems/find-pivot-index/
- https://leetcode.com/problems/running-sum-of-1d-array/
- https://leetcode.com/problems/range-sum-query-immutable/

- https://leetcode.com/problems/maximum-subarray/
- https://leetcode.com/problems/merge-intervals/
- https://leetcode.com/problems/maximum-average-subarray-i/

Medium Problems

Two Pointers

- https://leetcode.com/problems/two-sum-ii-input-array-is-sorted/
- https://leetcode.com/problems/container-with-most-water/

Sliding Window

- https://leetcode.com/problems/permutation-in-string/
- https://leetcode.com/problems/longest-repeating-character-replacement/

Prefix Sum

- https://leetcode.com/problems/subarray-sum-equals-k/

Kadane Pattern

- https://leetcode.com/problems/maximum-sum-circular-subarray/

Merge Intervals

- https://leetcode.com/problems/insert-interval/

Cyclic Sort Pattern

- https://leetcode.com/problems/find-the-duplicate-number/

Miscellaneous Array Patterns

- https://leetcode.com/problems/corporate-flight-bookings/
- https://leetcode.com/problems/majority-element-ii/

Hard Problems

- https://leetcode.com/problems/first-missing-positive/

On scope

This is a curated subset, not the full LeetCode tag list. It is sized so a learner can complete it within the phase timeframe while covering every pattern above at least once. Instructors can layer in additional problems per pattern from the same tag (Array, Two Pointers, Sliding Window, etc.) as stretch work.

Coding Challenges (Assessment Day)

- Maximum Subarray Sum (Kadane's Algorithm)
- Merge Intervals
- Subarray Sum Equals K
- Container With Most Water
- First Missing Positive

Expected Outcome

Students should be able to:

- Implement dynamic arrays from scratch, including resizing.
- Recognize common array patterns: traversal, frequency counting, prefix sum, difference array, two pointers, sliding window, Kadane's algorithm, merge intervals, and cyclic sort.
- Solve unseen array problems using pattern-based thinking rather than memorized solutions.
- Complete Easy and Medium array questions independently.